

MongoDB Practical Exam Preparation (SPPU BE AIDS - CL4)

This document contains complete step-by-step preparation for MongoDB Practical including:

- Installation
- Setup
- CRUD Operations
- Commands
- Viva Questions

1. Objective

Learn how to:

- Install MongoDB
- Create database and collection
- Perform CRUD operations
- Execute commands in Windows and Ubuntu

2. MongoDB Basics

MongoDB is a NoSQL document-oriented database.

Important Terms:

- Database → Contains collections
- Collection → Contains documents
- Document → JSON-like data record

Example Document:

```
{ name: "Asim", age: 22, branch: "AIDS" }
```

3. SQL vs MongoDB

SQL	MongoDB
Table	Collection
Row	Document
Column	Field
Fixed Schema	Flexible Schema

4. Windows Setup

Step 1 - Check MongoDB Installed

Open CMD and type:

```
mongosh
```

Step 2 - Download MongoDB

Download from:

<https://www.mongodb.com/try/download/community>

Step 3 - Install MongoDB

Tick:

- Install MongoDB as Service
- Install MongoDB Compass

Step 4 - Start MongoDB Service

Open Run and type:
services.msc

Step 5 - Open Mongo Shell

mongosh

5. Ubuntu Setup

Step 1 - Open Terminal

Ctrl + Alt + T

Step 2 - Check Installation

mongosh

Step 3 - Install MongoDB

```
sudo apt update  
sudo apt install mongodb
```

Step 4 - Start MongoDB

```
sudo systemctl start mongod
```

Step 5 - Enable MongoDB

```
sudo systemctl enable mongod
```

Step 6 - Check Status

```
sudo systemctl status mongod
```

Step 7 - Open Mongo Shell

mongosh

6. CRUD Operations

Create Database

```
use collegeDB
```

Insert Document

```
db.student.insertOne({name:"Asim", age:22, branch:"AIDS"})
```

Show Databases

```
show dbs
```

Show Collections

```
show collections
```

Read Documents

```
db.student.find()  
db.student.find().pretty()
```

Find One Document

```
db.student.findOne({name:"Asim"})
```

Update Document

```
db.student.updateOne({name:"Asim"}, {$set:{age:23}})
```

Delete Document

```
db.student.deleteOne({name:"Rahul"})
```

Exit MongoDB

```
exit
```

7. Important Viva Questions

Q. What is MongoDB?

Ans: MongoDB is a NoSQL document-oriented database.

Q. What is NoSQL?

Ans: Database that does not use tables and rows like relational databases.

Q. What is CRUD?

Ans: CRUD stands for Create, Read, Update and Delete.

Q. What is Collection?

Ans: Collection is a group of documents.

Q. What is Document?

Ans: Document is a JSON-like record storing actual data.

Q. Difference between SQL and MongoDB?

Ans: SQL uses tables while MongoDB uses collections.

8. Conclusion

MongoDB is a NoSQL database used to efficiently store and manage data in document format using CRUD operations.